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* Machine Learning :

- A field of study that gives computers the ability to learn without being explicitly programmed.

- 3 types
  ↳ Supervised
  ↳ Unsupervised
  ↳ Reinforcement

⇒ AI vs ML vs DL ⟶ 1/2 mark question

⇒ DL : uses neural network to learn using very large dataset (millions)
DL doesn't need human interaction. If model notices error it backtracks, then modifies necessary formula, then continues work.

* Supervised Learning :

- Type of machine learning where machines are trained using well labeled data and on that basis of data, machines predict the output.
  ↳ Classification problems (spam detection, sentiment analysis, fraud detection, predicting disease)
  ↳ Regression problems (predicting house/property/stock price)

* Unsupervised learning :

- The machine is trained using unlabeled data, and the machine predicts output without any supervision. Here the task of the machine is to group unsorted information according to similarities, patterns and differences without any prior training.

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↳ Association :

Discovering relationships between variables and data.

↳ Clustering :

Group data points according to their similarities.

* Reinforcement Learning :

A type of machine learning technique that enables an agent to learn in an interactive environment by trial and error using feedbacks from its actions and experiences. Each time an agent performs an action in its environment, a trainer may provide a reward or penalty to indicate the desirability of the resulting state.

* Steps in ML :

1. Collect data.
2. Process data : clean data by removing outlier.
3. Select a model to run.
4. Train : machine/model uses training data to make/create formula.
5. Test model
6. Tuning : if accuracy is less, we update the formula to improve performance.
7. Use for predicting.

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LINEAR REGRESSION.

X	Y	Mean X	Mean Y	Deviation X	Deviation Y	Product of Deviation DX * DY	Sum of product of Deviation	Square Deviation of X (DX^2)
8	10			-2	-3	6		4
10	13	10	13	0	0	0	12	0
12	16			3	3	6		4

Equation of Linear Regression :- $y = mx + b$

$$m = \frac{\text{Sum of product of deviation}}{\text{Sum of square deviation of } x} = \frac{12}{8} = 1.5$$

b = mean of y - (m * mean of x)

$= 13 - (1.5 * 10)$

$= -2$

$\Rightarrow$ For $x = 20$

$y = mx + b$

$= 1.5 * 20 + (-2)$

$= 28$

* Cost Function

- Mean squared error

$$J(w,b) = \frac{1}{n} \sum_{i=1}^{n} (\underbrace{\hat{y}_i}_{\text{predicted}} - \underbrace{y_i}_{\text{Actual}})^2$$

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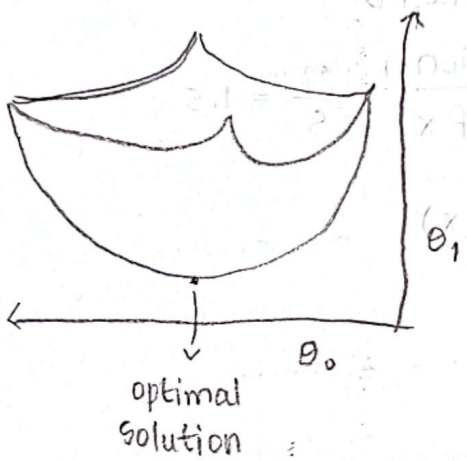
COST FUNCTION.

$y = mx + b$

$h_\theta(x) = \theta_0 + \theta_1 x$

cost function : $J(\theta_0, \theta_1) = \frac{1}{m} \sum_{i=1}^{m} (\hat{y}_i - y_i)^2$

$\theta_j = \theta_j - \underbrace{\alpha}_{\text{learning rate}} \frac{\delta}{\partial \theta_j} J(\theta_0, \theta_1)$ → slope



A 3D figure
2 variables θ_0 and θ_1

DECISION TREE.

- A machine makes decision based on conditions.

Ex :- Rain → not go to work
↓
no rain → go to work

- Algorithm used to make decision tree :- ID3

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Day	Outlook	Temp	Humidity	Wind	Play Tennis
0	Sunny	Hot	High	Weak	No
1	Sunny	Hot	High	Strong	No
2	Overcast	Hot	High	Weak	Yes
3	Rain	Mild	High	Weak	Yes
4	Rain	Cool	Normal	Weak	Yes
5	Rain	Cool	Normal	Strong	No
6	Overcast	Cool	Normal	Strong	Yes
7	Sunny	Mild	High	Weak	No
8	Sunny	Cool	Normal	Weak	Yes
9	Rain	Mild	Normal	Weak	Yes
10	Sunny	Mild	Normal	Strong	Yes
11	Overcast	Mild	High	Strong	Yes
12	Overcast	Hot	Normal	Weak	Yes
13	Rain	Mild	High	Strong	No

- We will calculate entropy to find the most important feature and make that as the root node.

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→ <u>Attribute</u> : Outlook

Values (outlook) : Sunny , Overcast , Rain

Entropy (S) = [9+, 5-] [+ : 'yes' outcome, - : 'No' outcome]

$$\text{Entropy}(S) = -\frac{9}{14}\log_2\frac{9}{14} - \frac{5}{14}\log_2\frac{5}{14}$$

$$= 0.94$$

$S_{sunny} \leftarrow [2+, 3-]$

$$\text{Entropy}(\text{Sunny}) = -\frac{2}{5}\log_2\frac{2}{5} - \frac{3}{5}\log_2\frac{3}{5} = 0.971$$

$S_{overcast} \leftarrow [4+, 0-]$

$$\text{Entropy}(\text{overcast}) = -\frac{4}{4}\log_2\frac{4}{4} - \frac{0}{4}\log_2\frac{0}{4} = 0$$

$S_{Rain} \leftarrow [3+, 2-]$

$$\text{Entropy}(\text{Rain}) = -\frac{3}{5}\log_2\frac{3}{5} - \frac{2}{5}\log_2\frac{2}{5} = 0.971$$

$$\text{Entropy}(S.\text{Outlook}) = \text{Entropy} - \sum \frac{|S_v|}{|S|}\text{Entropy}(S_v)$$

$$= \underbrace{0.94}_{\text{Total}} - \underbrace{\frac{5}{14}\times 0.971}_{\text{Sunny}} - \underbrace{\frac{4}{14}\times 0}_{\text{Overcast}} - \underbrace{\frac{5}{14}\times 0.971}_{\text{Rain}}$$

$$= 0.2464$$

→ <u>Attribute</u> : Temp

Entropy (S. Temp) = 0.0289

→ <u>Attribute</u> : Humidity

Entropy (S. Humidity) = 0.1516

→ <u>Attribute</u> : Wind

Entropy (S. Wind) = 0.0478

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Entropy is largest for outlook $\longrightarrow$ outlook is root node.

→ Next for sunny, we calculate entropy of other attributes.
No. of sunny outlook = 5

→ Values (temp) = Hot, Mild, Cool

$S_{Sunny} = [2+, 3-]$

Entropy (Sunny) = $-\frac{2}{5}\log_2\frac{2}{5} - \frac{3}{5}\log_2\frac{3}{5} = 0.97$

Similarly,

$S_{Hot} = [0+, 2-]$ Entropy = 0

$S_{Mild} = [1+, 1-]$ Entropy = 1

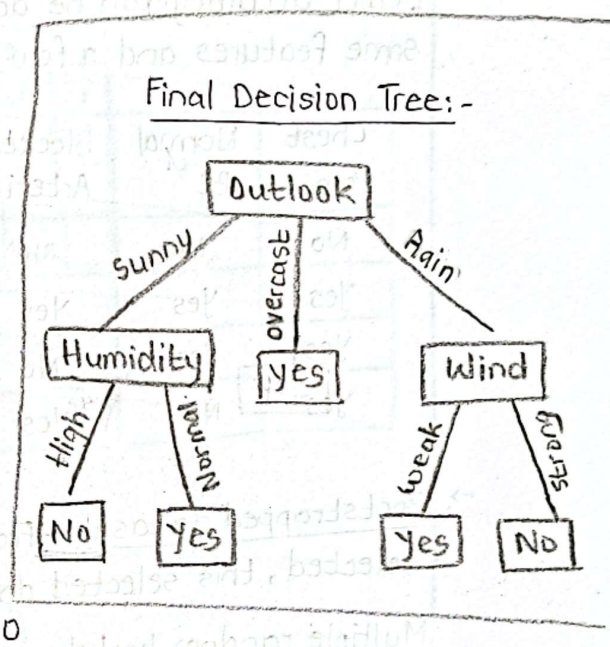
$S_{Cool} = [1+, 0-]$ Entropy = 0

Gain (Sunny, Temp) = $S - \sum \frac{|S_v|}{|S|}$ Entropy (S_v)

$= 0.97 - \frac{2}{5}\times 0 - \frac{2}{5}\times 1 - \frac{1}{5}\times 0$

$= 0.570$

Final Decision Tree:-



→ Gain (Sunny, Humidity) = 0.97

→ Gain (Sunny, Wind) = 0.0192

→ Gain is largest for Humidity, so it will become the next node/next question after sunny outlook.

→ Next, calculate gain of Temp, Humidity, Wind for all Rainy outlooks and select the one with largest entropy gain as the next node/next question after rain.

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RANDOM FOREST.

- A single decision may not be sufficient enough to make correct decision when dataset is large.

- Better accuracy can be achieved if we build multiple decision trees by taking some features and a few sample data for each tree.

Chest Pain	Normal BC	Blocked Arteries	Weight	Heart Disease
No	No	No	125	No
Yes	Yes	Yes	160	Yes
Yes	Yes	No	210	No
Yes	No	Yes	165	Yes

→ Bootstrapped dataset :- From original dataset, some random sample data are selected, this selected data is called bootstrapped data.

- Multiple random bootstrapped data are taken and multiple decision trees are built.

→ Out-of-Bag Data :- 1/3 of the data (original dataset) does not get selected while making bootstrapped dataset. This data is Out-of-Bag Data.

- This data samples are used for testing.

- Suppose there are 10 decision trees. After testing on all 10 trees, the final result will be whatever maximum decision is. [7 yes, 3 No → result = yes]

→ Bagging :- storing the verdicts of every decision tree.

→ Random forest can handle missing values by using proximity matrix.

Ex - Suppose 1st 3 entries have no missing values. The 4th entry has missing values in Blocked Arteries and weight.

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→ Initially fill the missing data by using average value or median value. Then run each sample data through all decision trees. Mark the samples that reach final verdict on the same leaf node and make proximity matrix.

→ <u>Proximity matrix</u> :-

	1	2	3	4 → data entry no.
1		2	1	1
2	2		1	1
3	1	1		8
4	1	1	8	

10 decision trees

	1	2	3	4
1		.2	.1	.1
2	.2		.1	.1
3	.1	.1		.8
4	.1	.1	.8	

Frequency for samples with no missing values

Yes = 1/3

No = 2/3

- Weighted frequency for 'yes' = $\frac{1}{3} \times 0.1 = 0.03$
- Weighted frequency for 'No' = $\frac{2}{3} \times 0.9 = 0.6$

So we place 'No' for Blocked Arteries

- Weighted average for sample 4 = $(125 \times 0.1) + (160 \times 0.1) + (210 \times 0.8)$

$= 198.5$

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